Inspectable Clinical Extraction from Epilepsy Letters

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***Abstract*—Clinical extraction results are difficult to interpret when model output, deterministic rules, repair, evidence checks, and scoring are reported as one method. We evaluate three methods on two epilepsy-letter tasks: current seizure-frequency labeling on the Gan 2026 synthetic benchmark and broad epilepsy phenotyping on ExECTv2. For each task we compare rules only, LLM only, and LLM with rules. Saved Gan holdout results are 364/450 Purist for the single-pass system and 379/450 for a multi-model comparison. On ExECT dev140, rules only reached 0.3548 strict item F1, the LLM only method reached 0.7393 clinical fact F1, and the combined method reached 0.9189. Three full200 model runs ranged from 0.8197 to 0.8566 clinical fact F1. Saved-output ablations show positive normalization contributions on both tasks, while the evidence check is score-inert on those representative replays. The selected evidence supports a reproducible component comparison with explicit data limits. It does not support independent clinical validation, strict ExECT benchmark reproduction, or a completed six-model result.**

***Index Terms*—clinical information extraction, epilepsy, large language models, reproducibility, evidence attribution**

I. Introduction

Epilepsy letters combine temporal reasoning, clinical terminology, medication regimens, investigation findings, and ambiguous current-versus-historical statements. Large language models (LLMs) can broaden extraction, but a model call can hide where a result came from. Deterministic systems are easier to inspect, yet often fail on linguistic variation and long-range clinical selection.

We compare rules-only, LLM-only, and LLM-with-rules methods. The same package contains a deep single-label task and a broad multi-entity task. This makes it possible to ask which stages transfer, which results depend on task-specific scoring, and where evidence is insufficient for a stronger claim.

The paper contributes: (1) selected results for three methods on two tasks; (2) explicit ownership of extraction, normalization, final formatting, repair, evidence checking, and scoring; (3) saved aggregate Gan holdout results and a three-model ExECT transfer study; and (4) reproducibility controls that pin prompts, scorers, splits, repair policies, runtime identifiers, dependencies, and file hashes.

II. Related Work

ExECT demonstrated structured epilepsy extraction from unstructured clinic letters and later work documented the corresponding annotation resource [1], [2]. Seizure-frequency extraction has also been studied with machine-reading and pretrained-language-model approaches [3], [4], while recent work has explored fine-tuned LLMs and reproducible synthetic clinical letters [5], [6]. Our focus is complementary: processing steps and data limits are part of the scientific method, and rules-only, LLM-only, and LLM-with-rules methods are compared without collapsing their different scoring and repair mechanisms.

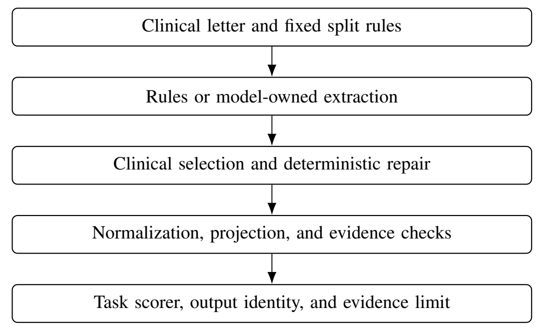

Fig. 1. Processing steps and their owners. Ownership remains explicit when a step is implemented by an LLM.

III. Methods

A. Tasks and Split Policy

Gan 2026 asks for one current seizure-frequency label per synthetic letter [6]. Validation750 is the row-inspectable development and replay split. Test450 is a locked holdout: only saved aggregate results are retained, and row-level holdout failures are not used for development.

ExECTv2 covers Diagnosis, SeizureFrequency, Prescription, and Investigations in the primary comparison, with additional entities retained in the deterministic all-nine reference [1], [2]. Dev140 is the row-inspectable development split. Full200 combines dev140 with held-out test60 and is reported only as a development-inclusive aggregate audit; it is not an independent holdout.

B. Compared Methods

For each task, the retained evidence index selects one rules-only, LLM-only, and LLM-with-rules run. Rules-only runs use no model calls. LLM-only runs put the clinical decision in one model program. The ExECT GEPA run is a negative development comparison. Combined runs keep model extraction distinct from deterministic selection, normalization, evidence checks, and scoring.

C. Scoring, Repair, and Attribution

Gan uses Purist and Pragmatic label scoring; Purist is the primary strict label score. ExECT's primary internal score is

TABLE I
SELECTED RESULTS FOR THREE METHODS ON TWO TASKS

Task	Method	Split	Selected result	Use and limit
ExECTv2	Rules only	dev140	Strict item F1 0.3548	Incomplete deterministic development reference.
ExECTv2	LLM only	dev140	Headline F1 0.7393	GEPA negative development comparator.
ExECTv2	LLM with rules	dev140	Clinical fact F1 0.9189	Current development reference.
Gan 2026	Rules only	validation750	697/750 Purist	Deterministic development comparator.
Gan 2026	LLM only	validation750	581/750 Purist	Single-pass development comparator.
Gan 2026	LLM with rules	validation750	661/748 rendered Purist	Single-pass event comparison.

TABLE II
GAN SAVED RESULTS ON LOCKED TEST450

System	Purist	Role
Single structured-event pass	364/450	Single-pass system
V12 multi-trace reviewer	379/450	Multi-model comparison

de-duplicated `clinical_headline` recovery across four families. Phrase, CUI, evidence-valid, and full-attribute companions are reported separately. `clinical_headline` is not presented as reproduction of the published strict ExECT benchmark.

Raw model selection, JSON or format repair, source-exact evidence repair, deterministic semantic repair, projection, and final scoring are separate stages. Model-specific transport or output-shape adapters do not become model quality evidence. Semantic repairs retain a named deterministic owner and must be ablated or otherwise tested before supporting a causal interpretation.

D. Recorded Reference Version

The retained evidence index records implementation commit `46562134`, the dependency declaration and lock, prompts, scorers, split definitions, repair policies, model policy, split runbook, and CI workflow. A semantic prompt, scorer, split, repair, runtime-route, or component-graph change requires a new recorded version and complete replay. This rule does not authorize model calls.

IV. RESULTS

A. Selected Method Comparison

Table I is the minimum selected scientific comparison. Scores are not compared across tasks because the datasets, outputs, and scorers differ.

All six runs replay from current code and selected outputs without model calls. The ExECT combined-method replay also returns evidence-valid F1 0.8913. Its current benchmark/CUI companion is 0.4791 versus 0.4729 in the saved run; this companion-scorer drift is open and does not affect the reproduced 0.9189 headline result.

TABLE III
EXECT SAME-CORE FULL200 AGGREGATE `CLINICAL_HEADLINE`

Model condition	Overall	Diagnosis	SF	Prescription	Investigations	Call/parse failures
DeepSeek chat	0.8566	0.8708	0.7602	0.8926	0.9091	0/1
GPT-4.1-mini	0.8356	0.8397	0.7525	0.8926	0.8563	0/0
Qwen 3.6 35B repair v02	0.8197	0.8307	0.7020	0.8926	0.8503	0/0

TABLE IV
SAVED-OUTPUT COMPONENT ABLATION

Component	ExECT dev140	Gan validation750
Normalization/dictionary	+0.0389	+0.0293
Evidence validation	0.0000	0.0000

TABLE V
RETAINED RELIABILITY EVIDENCE AND LIMITS

Subject	Retained result	Boundary
ExECT evidence	Hybrid dev140 minimum exact evidence rate 1.0000	Development reference.
ExECT calibration	Brier 0.2225; base-rate Brier 0.2340; ECE 0.0587	Full200 aggregate internal scoring-rule result.
Gan grounding	Architecture-specific validation grounding packages retained	Metrics are not uniform across families.
Reproducibility	1,157 tests, Ruff, mypy, hashes, split restrictions, and six-run replay pass on Python 3.11	Engineering verification, not clinical validation.

B. Gan Locked Holdout Evidence

Both rows in Table II are saved aggregate evidence. The 15-row quality difference does not yet support an efficiency conclusion. A matched table of calls, tokens, cost, latency, hardware, and cache policy remains open.

C. ExECT Same-Core Model Evidence

The retained model-transfer package holds the two-call no-SF-adjudicator component graph fixed while changing the runtime model. Table III is a full200 development-inclusive aggregate read; no test60 row-level analysis is used.

These results apply to the three named runtime conditions. They are not a strict same-prompt panel: the retained GPT condition used temperature 0.3, and Qwen used a compact prompt and output-contract repair. The requested six-model comparison remains incomplete, with three missing runtime conditions requiring predeclaration.

D. Cross-Task Components and Reliability

The zero score delta in Table IV does not show that evidence validation is unnecessary. A filter may be essential on malformed or hallucinated outputs even when all selected reference predictions already satisfy it. Rejection and repair tests, not aggregate F1 alone, carry that part of the evidence.

Model-reported confidence remains unused, and no low-burden review-routing policy is adopted. The retained evidence index does not include evidence for a cross-task over-reading claim.

V. DISCUSSION

The reduced system can reproduce six selected runs while keeping component ownership and data limits explicit. The combined ExECT method is substantially stronger than the retained ExECT rules-only and LLM-only development references on `clinical_headline`, but the three runs do not share a strict published benchmark score. Deterministic published-metric engineering remains open.

The model-transfer result shows that the component graph runs with three qualitatively different runtime models. DeepSeek leads the retained full200 aggregate, while Qwen trails most clearly on SeizureFrequency. Because runtime conditions are asymmetric and the panel is only three of six, this is evidence that applies only to these three runs, not a universal model claim.

The cross-task ablation suggests that normalization is useful in both tasks. The exact-evidence check's zero F1 delta shows why aggregate score changes are insufficient: rejection, source attribution, and repair need direct tests.

VI. Limitations

Gan test450 contains saved aggregate evidence only and cannot support row-level tuning. ExECT full200 includes dev140 and is not an independent holdout. `clinical_headline` is a clinical fact score, not the published strict benchmark. The ExECT benchmark/CUI companion has small current-code replay drift. The model panel is three of six and uses asymmetric runtime conditions. Model-reported confidence and low-burden review routing are not validated. Annotation-quality findings are internally adjudicated; unqualified clinical validity would require independent domain review. The current retained evidence does not support a cross-task over-reading claim.

VII. Conclusion

An inspectable clinical extraction system can preserve performance evidence and the source needed to interpret it. The selected results support three methods on two tasks, saved Gan holdout results, and three named ExECT model runs. They do not yet support strict ExECT benchmark reproduction, a six-model conclusion, or independent clinical validation. Those boundaries are part of the result, not footnotes to it.

Reproducibility Statement

The machine-readable retained evidence index records the six runs, selected files and hashes, policy versions, and no-call replay expectations. The Markdown manuscript and this IEEE source use only selected evidence and claims permitted by the paper claim status.

References

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